Whitney Learning Compass
From Curriculum Goals to Hands-On Discovery

Catapult · Learning Plan

SELECTED OFFICIAL PROGRAM · READ-ONLY FACTS

Program Requirements Met

Catapult

Students experiment with a rubber-band-powered catapult, varying band count and the position of the supplied ping-pong ball to explore force and motion.

PUBLISHED GRADES	SUBJECTS	PUBLISHED FEE
3, 4, 5, 6, 7+	Design · Energy Transfer · Engineering · Force and Motion · History	$14.00 per student

CLASS PROGRAM FEE	WORKSHOP DURATION	BOOKING
USD 336.00	90 minutes	Not confirmed

USD assumed for this Connecticut prototype; not explicitly published by the museum.

Program fees exclude transportation, meals and unverified additional charges.

Your class: Grade(s) 4 · 24 students

Learning goal: I want my fourth-grade students to explore force, motion, and engineering design through hands-on experiments.

Limits: USD 400 program-fee budget · 90 minutes maximum

TEACHER-EDITED SUGGESTIONS

AI-generated editable suggestions for educator review; not museum curriculum or endorsement. Edits stay in this browser tab and do not trigger model calls. Save a printed copy before closing or refreshing.

BEFORE

Prepare & Wonder

Build curiosity before the workshop.

Learning objectives

Students will identify how changing the number of rubber bands or the ping-pong ball's position relates to the catapult's motion, using observations from the program. Use an actual visit observation as evidence.

Students will describe one step in an engineering-design process by comparing a prediction with observed results and explaining a possible reason for any difference.

Classroom preparation

Invite students to recall examples of pushes and pulls and how they can change an object's motion.

Prepare a simple paper-and-pencil recording sheet with columns for trial or setup, prediction, actual observation, and possible explanation; leave observation fields blank until observations occur.

Review that any catapult use and handling during the program follow museum staff directions.

Background concepts

A force is a push or pull that can change an object's motion.

The catapult activity varies band count and the position of the supplied ping-pong ball; students can compare what they observe across those changes.

Engineering design involves making a prediction, observing a result, and using evidence to consider what to change or explain.

Inquiry questions

How might changing the number of bands affect the catapult's motion?

How might changing the ping-pong ball's position affect what happens?

What evidence would help you compare two setups fairly?

DURING

Explore & Observe

Follow staff directions. Collect evidence, not invented results.

Observation prompts

If museum staff permit, attend to how the catapult's motion appears to change when the band count or ball position changes; record only what is actually observed.

Use a blank table: setup/change | prediction | actual observation | possible explanation. Leave the actual-observation column empty until the event is observed.

Follow museum staff directions for all catapult handling and procedures; do not adjust or operate equipment independently.

Hypothesis & prediction questions

Before a staff-directed comparison, what do you predict may happen if the band count changes? Label this as a prediction, not an observation.

What do you predict may happen if the ball's position changes? What makes you think so?

Critical-thinking prompts

Which factor changed between the setups you observed, and what appeared to stay the same?

Does the observed result support your prediction, differ from it, or leave the question open? What specific observation informs your thinking?

What other factor might affect the motion, and what would you need to observe before drawing a conclusion?

Evidence collection

Record the setup and the directly observed result in words or a simple sketch; do not fill in results before observing them.

Keep predictions separate from observed evidence, and note when an observation is unavailable or unclear rather than guessing.

Ask museum staff before recording or comparing any details not included in the staff-directed activity.

AFTER

Reflect & Extend

Connect actual observations to your learning goals.

Reflection & discussion

Which actual observation most helped you think about force and motion? Explain how it connects to the setup.

Where did an observation support or challenge your prediction? What evidence led you to that view?

What would you still want to observe before making a stronger claim about how a setup affects motion?

Follow-up classroom activity

On paper, have students choose one observed comparison and make a labeled sketch of the setup, then write a claim about motion supported by an actual recorded observation. If no observation record is available, use the blank table to plan what they would look for next; do not supply invented results.

Learning assessment

Objective 1: Review each student's explanation for a connection between a changed setup feature (band count or ball position) and motion, supported by an actual observation. If none is available, assess the student's ability to state what evidence would be needed instead.

Objective 2: Check whether the student separates prediction from observed result and uses the observation to explain a possible outcome or difference; do not score hypothetical outcomes as empirical evidence.

Optional extension

Students can add a paper-and-pencil design reflection: identify one feature they would want to compare next, state a clearly labeled prediction, and list the actual observation that would be needed to evaluate it.

Teacher notes & actual observations

Leave observations blank until the visit occurs. Label predictions as predictions; do not treat them as collected evidence.

No observations recorded. To be completed by the educator after actual observation.

Questions to Confirm with the Museum

Eligibility does not establish bookability.

- What accessibility accommodations can you arrange?
- Are there any additional charges beyond the published per-student program fee?
- What dates are available for this program?
- What booking conditions and reservation steps apply?
- Can you accommodate our class of 24 students?
- Are there any special delivery or location arrangements to confirm?

Safety & Important Notes

Educator review required. Follow museum staff procedures; do not independently operate equipment or substitute projectiles. Use only approved activities and materials.

The museum's safety statement is conditional on using its supplied ping-pong ball; this record does not establish a general safety guarantee.

Manual live official-page review completed on 2026-10-21. Published facts are verified as displayed; booking, delivery, and future price validity are not established.

Record actual observations. Clearly label predictions, and leave observation notes blank until evidence has been collected.

Sources & Verification

https://www.eliwhitney.org/catapult

Reviewed: 2026-10-21. Program-specific standards: Not published.

/description · program_page

/grade_levels · program_page

/subjects · program_page

Manual live official-page review completed on 2026-10-21. Published facts are verified as displayed; booking, delivery, and future price validity are not established.

The museum's safety statement is conditional on using its supplied ping-pong ball; this record does not establish a general safety guarantee.

USD assumed for this Connecticut prototype; not explicitly published by the museum.

Teacher review acknowledged. Teaching suggestions are not official museum curriculum. No experiment or student outcome is claimed to have occurred.

Independent educational prototype based on publicly available Eli Whitney Museum & Workshop program information. Not affiliated with, endorsed by, or operated by the museum.